

Technology Selection

Technology Area	Selected	Why Selected
Frontend	React	Flexible component-based UI
Backend	Python+FastAPI Python libraries • python-docx - Word document extraction and structure • PyMuPDF - PDF processing • difflib - Section/content comparison • hashlib - Exact duplicate detection • re - ISO template/format validation	Strong document processing and AI/NLP potential
Database	MongoDB	Flexible schema for document/version data
Automation	Power Automate	Strong SharePoint/Microsoft 365 integration
Repository	SharePoint	Explicitly required by SRS
API Integration	Microsoft Graph API	Designed for Microsoft 365/SharePoint integration
Host	Microsoft Azure	Provides a scalable cloud environment for hosting the React frontend and FastAPI backend while offering strong compatibility with the Microsoft ecosystem used by the system, including SharePoint, Microsoft Graph API and Power Automate.